H3. The OrsD (PF12013) gene set: what is the excess of low-dS duplicates?

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Contents

H3.1 The excess comes from a few genomes 1
H3.2 The excess is assembly (haplotype) duplication 2
H3.3 The recent carriers are gene fragments 2
H3.4 The recent copies are tandem arrays 3
H3.5 The recent copies sit next to transposable elements 3
H3.6 The family is a helicase-domain gene family with dispersed recent copies 4
Figure 5
Reproduce 5

OrsD is the Pfam name in the database. Its function and family description were not verified (a web search returned no entry). This report tests what the data show about the gene set. Method: `scripts/domain_geneset.py`, `domain_te_context.py`, `domain_deepdive.py`.

Status: interim. All numbers come from the 7,384-genome interim tables (yn00 dS). They will change when the final tables are published.

Starting observation. Among carriers of PF12013 that sit in a gated pair, 48.5% are at dS <= 0.1. Among all paralog genes the share is 22.1%. In the clean genome set the odds ratio against other paralogs is 6.8 (95% CI 6.4 to 7.2; length- and genome-stratified Mantel-Haenszel; `tables/gene_enrich_recent_pf_clean.tsv`). The gene set has 44,086 proteins, of which 37,234 are in no gated pair.

bin	domain_genes	all_paralog_genes	share_domain_in_pairs	share_all
H1	2407	3.43e+06	0.351	0.176
MID	1589	5.4e+06	0.232	0.278
OLD	825	6.22e+06	0.12	0.32
Y	1114	3.51e+06	0.163	0.181
Z	917	8.69e+05	0.134	0.0447
not in a gated pair	37234	0		0

H3.1 The excess comes from a few genomes

Hypothesis. A small number of genomes carry a large family of near-identical OrsD copies and drive the pooled result.

It would count against the hypothesis if: most recent carriers sit in the top handful of genomes, or few genomes show carrier excess on their own.

Test. Share of recent carriers in the top 10 genomes; number of genomes with recent carriers; per-genome Fisher test of carrier recent share against the same genome’s other paralogs, in genomes with BUSCO duplication < 20% and >= 5 carriers in pairs. Verdict rule: REJECTED if top-10 share < 30%, >= 100 genomes, and > 30% of testable genomes show excess.

Result. Top 10 genomes hold 14.9% of recent carriers. 683 genomes have at least one, 206 at least five. Of 422 testable low-duplication genomes, 41.9% show carrier excess (p < 0.01). Median recent share: carriers 0.50, all paralogs 0.135.

genome	class	busco_duplicated_pct	paralog_genes	recent_share_all_paralogs	carriers_in_pairs	carriers_recent	fisher_OR	p
Humicolopsis_cephalosporioides_CH-1.1	Leotiomycetes	0.7	4453	0.396	116	90	5.5	4.14e-17
Puccinia_graminis_21-0	Pucciniomycetes	81.3	31995	0.721	83	79	7.68	1.1e-07
Lizonia_empirigonia_CBS542.76	Dothideomycetes	0.7	1783	0.46	70	68	43.4	1.88e-21
Histoplasma_capsulatum_WU24	Eurotiomycetes	0	1890	0.481	47	47	inf	6.08e-16
Laburnicola_sp._JP-R_1_JF_R_19	Dothideomycetes	96.7	22479	0.813	52	46	1.76	0.215
Aspergillus_spinosus_NRRL_5034	Eurotiomycetes	0.8	1942	0.193	43	34	17.3	1.51e-17
Tricholoma_sulphureum	Agaricomycetes	0.9	13980	0.517	34	34	inf	1.95e-10
Cercospora_sojina_CCC_172-09-ARG_09_172	Dothideomycetes	0.1	1771	0.215	34	33	132	8.46e-22
Pseudopyrenochaeta_lycopersici_CRA-PAV_ER_1518	Dothideomycetes	0.3	4924	0.407	33	32	47.4	5.48e-12
Laburnicola_rhizohalophila_JP-R2018DY-8A	Dothideomycetes	89.4	26453	0.873	35	31	1.13	1

Verdict: REJECTED. Several of the largest cases come from genomes with small total paralog sets, where one family is a large part of the set.

H3.2 The excess is assembly (haplotype) duplication

Hypothesis. Recent carrier pairs are allelic copies in dikaryotic or doubled assemblies, so the excess disappears once genomes with high BUSCO duplication are removed.

It would count against the hypothesis if: the enrichment persists in genomes with BUSCO duplication < 20%.

Test. Compare the clean-set odds ratio with the all-genomes odds ratio, and carrier against all-paralog recent share in high-duplication genomes.

Result. The clean-set enrichment is OR 6.8 (all genomes: 5.1; `tables/gene_enrich_recent_pf_all.tsv`). In genomes with BUSCO duplication >= 40% (n = 21 testable), recent share is 0.74 for all paralogs and 0.83 for carriers. Examples: *Puccinia graminis* 21-0 (BUSCO dup 81%): 79 of 83 carriers recent against 72% of all paralogs; *Melampsora americana* (60%): carriers depleted (OR 0.17).

Verdict: REJECTED as the main explanation. Assembly duplication contributes in a few rust genomes, but the excess persists in low-duplication genomes.

H3.3 The recent carriers are gene fragments

Hypothesis. Recent carriers are short partial models that carry part of the domain.

It would count against the hypothesis if: carriers are long and carry most of the Pfam model.

Test. Protein length and Pfam model coverage (hit length / model length) of carriers by bin.

Result.	bin	n	med_len	med_domain_cov	frac_cov_ge_0_8
H1	2407	1.01e+03	0.956	0.885	0.0158
MID	1589	759	0.956	0.88	0.0302
OLD	825	920	0.956	0.939	0.0194
Y	1114	695	0.956	0.865	0.0305
Z	917	1.28e+03	0.974	0.936	0.0338
not in a gated pair	37234	607	0.965	0.91	0.0405

Verdict: REJECTED. Median carrier length is about 1,000 aa in the youngest bins and median model coverage 0.96. This differs from the general result in H2.2. The Sorochytrium genome does have short copies of 284 to 711 aa without a helicase domain, so fragments exist in some genomes.

H3.4 The recent copies are tandem arrays

Hypothesis. Recent carrier-carrier pairs lie next to each other on the same contig.

It would count against the hypothesis if: most pairs are on different contigs or far apart.

Test. For recent pairs in which both genes carry the domain: share on the same contig, distance, share within 20 kb.

Result. 3,663 carrier-carrier recent pairs. Same contig: 0.067. Same contig and within 20 kb: 0.016. Median distance on the same contig: 296,214 bp.

Verdict: REJECTED. Contigs are fragments of a chromosome in most assemblies, so different contigs can still be close in the real genome; the rate of same-contig pairs is a lower bound on local clustering.

H3.5 The recent copies sit next to transposable elements

Hypothesis. Recent carriers have TE-domain genes within 25 kb more often than matched control genes.

It would count against the hypothesis if: no excess of nearby TE-domain genes, with a confidence interval that excludes a large effect.

Test. TE marker: genes carrying one of 15 TE Pfam domains. Each recent carrier is compared with up to 5 control genes from the same genome and protein-length class (Mantel-Haenszel odds ratio; strata genome x length class). Verdict rule: SUPPORTED if the interval excludes 1 and OR > 1; NOT SUPPORTED if the interval includes 1.

Result.	grp	n	genomes	self_TE_domain	TE_gene_within_10kb	TE_gene_within_25kb
control	34260	1436	0.0007881	0.01141	0.02957	7.903e+05
older	3528	1171	0	0.02857	0.05714	3.742e+05
recent	3324	683	0.0003008	0.01235	0.02469	9.396e+05

group	MH_OR_TE_within_25kb	ci_lo	ci_hi	strata
recent	1.65	0.841	3.25	219
older	1.34	0.52	3.47	121
recent, BUSCO dup<20% genomes	1.84	0.845	3.99	193

Verdict: INCONCLUSIVE (no evidence for; low power). Recent carriers: OR 1.65 (95% CI 0.84 to 3.25). Only about 3% of control genes have a TE-domain neighbour within 25 kb, so the test has low power and the interval is wide. The marker is crude (15 domains). A repeat-annotation test

(RepeatMasker or EarlGrey) on these genomes was not possible because no genome-wide repeat annotation exists for them in the tables.

H3.6 The family is a helicase-domain gene family with dispersed recent copies

Hypothesis. Recent carriers share a common architecture (helicase domains) and form few gene families within a genome.

It would count against the hypothesis if: carriers have varied domain architectures and no dominant partner domain.

Test. Co-occurring Pfam domains on recent carriers; in the *Sorochytrium milnesiophthora* genome (Blastocladiomycetes; 23 carriers, 21 recent) the gene list, contig positions and gene family.

Result.	acc	pname	proteins
PF00270	DEAD	1433	0.431
PF00271	Helicase_C	1215	0.366
PF04851	ResIII	657	0.198
PF07652	Flavi_DEAD	14	0.00421
PF11951	Fungal_trans_2	7	0.00211
PF24883	NPHP3_N	5	0.0015

Sorochytrium: 23 carriers on 17 contigs; 1 gene family; median length 711 aa; 11 of 23 carriers start within 10 kb of the contig start.

protein_id	chrom	start	length	ds	n_recent_pairs	other_pfam
F5C4F326_001409-T1	JAVQLU010000064.1	70	1413	0.0203	5	PF04851:ResIII,PF00270
F5C4F326_001413-T1	JAVQLU010000064.1	23551	1657	0.00881	6	PF00270:DEAD,PF00271
F5C4F326_001998-T1	JAVQLU010000036.1	83	498	0.0323	2	
F5C4F326_002769-T1	JAVQLU010000056.1	558624	1672	0.113	0	PF00270:DEAD,PF00271
F5C4F326_002774-T1	JAVQLU010000009.1	4298	1711	0.00314	2	PF00270:DEAD,PF00271
F5C4F326_003613-T1	JAVQLU010000012.1	498540	284	0	2	
F5C4F326_003616-T1	JAVQLU010000012.1	514698	432	0	2	
F5C4F326_003618-T1	JAVQLU010000012.1	519760	354	0	4	
F5C4F326_003842-T1	JAVQLU010000014.1	4379	2577	0.00314	1	PF13959:CTE_SPB4,PF0
F5C4F326_005708-T1	JAVQLU010000023.1	393484	1146	0.0136	4	
F5C4F326_005709-T1	JAVQLU010000023.1	398040	1782	0.0461	4	PF04851:ResIII,PF00270
F5C4F326_005833-T1	JAVQLU010000023.1	374314	1671	0.00881	5	PF04851:ResIII,PF00270

Verdict: SUPPORTED (descriptive). DEAD, Helicase_C and ResIII co-occur on 43%, 37% and 20% of recent carriers, and 54% of carriers have no other Pfam domain. This describes the set. It does not show that the copies are mobile. Verifying the family would need sequence similarity to known families and repeat annotation.

Figure

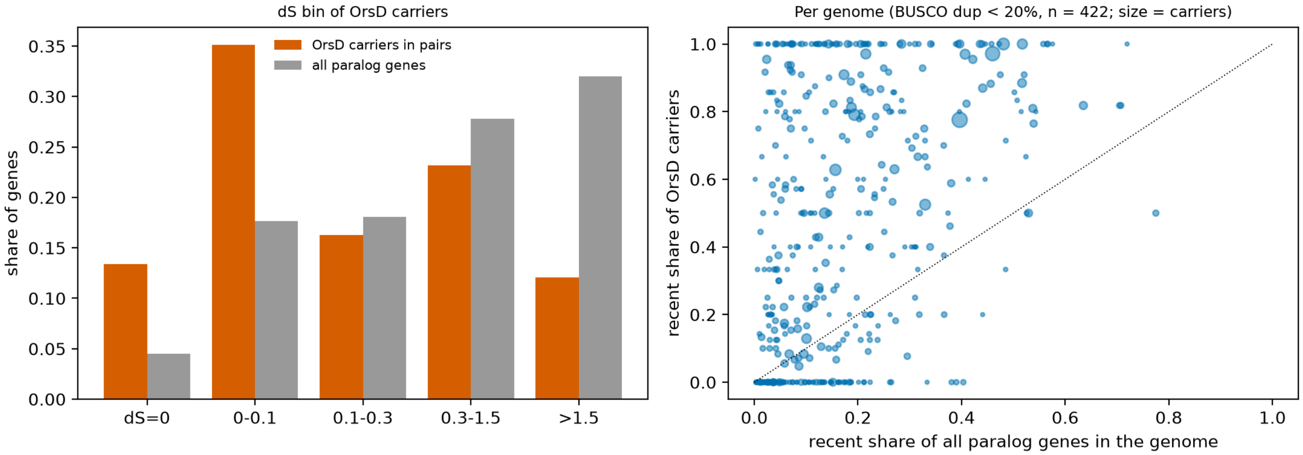

Figure 1: OrsD carriers

Reproduce

`scripts/domain_geneset.py PF12013 OrsD`, `scripts/domain_te_context.py PF12013 OrsD`, `scripts/domain_deepdive.py PF12013 OrsD`, `scripts/make_hypothesis_reports.py`.